Check for updates

ASME Journal of Engineering for Gas Turbines and Power
Online journal at:
https://asmedigitalcollection.asme.org/gasturbinespower

# On the Optimum Distribution of Multiple Helmholtz Resonators for Annular Combustors

**Hongwei Chen**
The Department of Mechanics and Aerospace Engineering,
Southern University of Science and Technology,
Shenzhen, Guangdong 518055, China

**Liming Yin**
The Department of Mechanics and Aerospace Engineering,
Southern University of Science and Technology,
Shenzhen, Guangdong 518055, China

**Xiaopei Liu**
Shanghai Electric Gas Turbine Co., Ltd.,
Shanghai 200240, China

**Mingmin Chen**
College of Power and Energy Engineering,
Harbin Engineering University,
Harbin, Heilongjiang 150001, China

**Dong Yang**[1]
The Department of Mechanics and Aerospace Engineering,
Southern University of Science and Technology,
Shenzhen, Guangdong 518055, China
e-mail: yangd3@sustech.edu.cn

*The combustors of many modern land-based gas turbines and aero-engines are annular. This kind of combustors often suffer from thermoacoustic oscillations, with the occurrence of mostly the circumferential first-order, second-order, and third-order oscillation modes in the annular combustion chamber. This could cause severe pressure oscillations in the combustor, which may increase noise and $NO_x$ emissions, affect the safe operation of the engine and even cause irreversible damages to the structure. To solve this problem, passive control methods are widely used—adding passive acoustic dampers such as Helmholtz resonators (HRs). Depending on the type and number of HRs, and the possible positions over the circumference to install them, there could easily have millions, or even billions, of possible arrangement patterns. Finding a good design is the key to solve this problem. In this paper, we perform a theoretical and numerical study of an annular combustor installed with multiple types of HRs over the circumference. First, a simple annular duct with arbitrary distributions of these HRs is studied analytically by solving the nonlinear eigenvalue problem of a one-dimensional network model. Based on the results of the analytical method, the impact of the HRs on the acoustic modes of the combustion chamber is studied and the optimum arrangement of multiple resonators is obtained. This arrangement usually gives a null (or small) mode splitting strength and a good damping effect. Finally, we apply the optimum arrangement to damp the thermoacoustic modes that have been captured by numerical simulation for a real annular combustor. We use numerical simulations based on solving the three-dimensional Helmholtz equation in COMSOL to verify the feasibility of the optimum arrangement.* [DOI: 10.1115/1.4066354]

*Keywords: Helmholtz resonator, optimum distribution, thermoacoustic instability, annular combustor*

## 1 Introduction

In order to achieve low $NO_x$ emission, modern aero-engines and land-based gas turbines often use the concept of lean premixed combustion. This significantly increases the risk of thermoacoustic instabilities in the combustion system. Many of these combustors are annular in shape, with the circumference being much longer than or of the same order of magnitude as the longitudinal length. As thermoacoustic instabilities usually happen in the low-frequency region, this results in thermoacoustics modes in annular combustors to be usually associated with the first 1–3 azimuthal acoustic modes [1–4].

Acoustic damping devices, such as Helmholtz resonators (HRs), quarter-wave tubes, etc., are widely used to damp thermoacoustic oscillations. They suppress thermoacoustic oscillations by increasing the acoustic energy dissipation of the combustion process and breaking the phase angle between pressure and heat release [5]. Many studies have been performed to improve the acoustic absorption performance of the HR itself, or to better model its influence on the acoustics inside the combustor. For example, the neck length, diameter, and the cavity volume have been considered to increase the sound absorption bandwidth [6–8]. New types of resonators, such as the one with a cavity accommodating multiple secondary necks connecting each other has also been studied [9]. When the resonator is installed in the combustion chamber whose temperature is high, the effect of introducing cooling air into the HR has been studied. A new model of the acoustic-entropy wave relationship inside the combustor was developed [10] to consider the influence of the temperature difference. Betz et al. [11] used different methods to calculate the damping rate and compared it with the experimental results. Based on a realistic annular combustor, the damping effect was calculated and the effect of HRs on the thermoacoustic modes was experimentally studied [12]. Zahn et al. [13] solved the linearized Euler equations to predict the stability of the combustion chamber after the addition of HR. Mensah and Moeck [14] proposed a Helmholtz solver combined with the adjoint method, and used it to study the effect of installing multiple resonators on the stability of the combustion chamber. By combining an adjoint-based sensitivity analysis and a gradient-based optimization method, Yang et al. [15] utilized a two-dimensional low-order network mode to optimize multiple HRs targeting multiple thermoacoustic modes.

Turbo Expo, June 24–28, 2024. GT2024.
[1]Corresponding author.
Manuscript received July 12, 2024; final manuscript received July 25, 2024; published online September 26, 2024. Editor: Jerzy T. Sawicki.


Copyright © 2025 by ASME; reuse license CC-BY 4.0

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

When the HRs are attached to the combustion chamber, they could be placed at the inlet, middle, and outlet of the combustion chamber. Experiments have shown that the best damping effect could usually be obtained when they are placed at the inlet of the combustion chamber [9,11]. As the first several azimuthal modes are the most important target modes, when the longitudinal location is fixed, the number and distribution patterns of the HRs arranged over the circumference can make a significant difference and have also been widely studied. When more HRs are used, the damping performance is usually better. It has also been known that the detailed distribution pattern could significantly change their damping performance—when the dampers are arranged in different patterns, i.e., symmetrically or asymmetrically, in the circumferential direction, the damping performance could be rather different [9,16–20]. However, at least two difficulties arise when one wants to systematically study this: (i) when tens of HRs with several types are arranged over the circumference, millions to billions of possible different patterns could be easily resulted in, exhausting these possibilities to obtain the optimum distribution is difficult; (ii) in many cases, especially for the optimum distribution case, modes may merge or nearly merge, making accurately resolving the nonlinear eigenvalue problem especially challenging.

Regarding the effect of the distribution of different burners on the splitting and stability of the azimuthal thermoacoustic modes, Bauerheim et al. [21] have developed an analytical model based on a Taylor expansion of the nonlinear eigenvalue problem around the acoustic modes of the combustion chamber. In this model, the effect of mixing different burners on the splitting and stability of the azimuthal thermoacoustic modes can be explicitly obtained. This provides a very potential capability to study the effect of multiple dampers and possibly to find their optimum distributions over the circumference.

Based on previous studies, the important influence of damper arrangements on azimuthal thermoacoustic oscillations has been recognized and it can be seen that a systematic study on the dampers' optimum distribution over the circumference could be very valuable. This paper focuses on this topic. We extend the theoretical model from Ref. [21] to systematically study the optimum circumferential arrangement of HRs given their specific types and number. The three-dimensional (3D) FEM Helmholtz solver from COMSOL is used to simulate the annular combustion system of a realistic gas turbine from Shanghai Electric (Shanghai, China) to verify the effect of the optimum arrangements.

## 2 Theoretical Models

Following a similar line as the annular network reduction model from Ref. [21], an annular combustion chamber is uniformly divided over its circumference into $N$ parts, each with a central angle $2\pi/N$ and an arc length of $2\pi R/N$. An HR with impedance $Z_i$ ($i = 1, 2, \ldots, N$) is attached to each segment, as shown in Fig. 1(*a*). The theoretical model to determine the eigenvalues of this problem was derived in Ref. [23]. To aid the readers, we briefly introduce the main steps.

**2.1 Network Model.** Assuming that the axial length of the combustor is much shorter than its circumference, so for the low frequency thermoacoustic mode, only acoustic wave propagation on the circumferential direction needs to be considered. The circumferential length between each two HRs is $X_c = R\theta = 2L_c/N$, as shown in Fig. 1. By using jump conditions at the $i$th HR (between the perturbations at $i+1/2$ and $i+1$), a transfer matrix can be expressed as

$$\begin{bmatrix} p' \\ \rho_0 c_0 u' \end{bmatrix}_{i+1} = \begin{bmatrix} 1 & 0 \\ 2j\Gamma_i & 1 \end{bmatrix} \begin{bmatrix} p' \\ \rho_0 c_0 u' \end{bmatrix}_{i+1/2} \tag{1}$$

where $j$ is the imaginary unit, $\Gamma_i$ is

$$\Gamma_i = -\frac{j}{2}\frac{S_i}{S_c Z_i} \tag{2}$$

$Z_i$ is the impedance, $S_i$ is the neck cross-sectional area of the $i$th HR, and $S_c$ is the longitudinal cross-sectional area of the annular combustion chamber. Between the $i$ and the $i+1$ HR, the propagation of azimuthal waves can be expressed by a transfer matrix

$$\begin{bmatrix} q^+ \\ q^- \end{bmatrix}_{i+1} = \begin{bmatrix} W & 0 \\ 0 & \frac{1}{W} \end{bmatrix} \begin{bmatrix} q^+ \\ q^- \end{bmatrix}_i \tag{3}$$

where $q^\pm = p' \pm \rho_0 c_0 u'$, $W = e^{2jkL_c/N}$, and the wavenumber $k = \omega/c_0$.

Considering the periodic boundary condition over the circumference annular network reduction methodology furnishes an implicit analytical dispersion relation for the angular frequency $\omega$

$$\det\left(\prod_{i=1}^{N} \boldsymbol{M}_i - \boldsymbol{I}_d\right) = 0 \tag{4}$$

where $\boldsymbol{I}_d$ is the $2 \times 2$ identity matrix.

For annular chambers without HRs, the eigenfrequencies are real: $f = rc_0/2L_c$, $r = 1, 2, 3 \cdots$. When HRs are installed on the annular chamber, they can be seen as a minor perturbation to the combustion chamber's acoustic modes, given that $|\Gamma_i| \ll 1$. Using Taylor expansion of the entire system around $f = rc_0/2/L_c$ up to the second-order, yielding

$$\begin{aligned} &\det\left(\prod_{i=1}^{N} \boldsymbol{M}_i - \boldsymbol{I}_d\right) \\ &\approx -\frac{W^{2N} - 2W^N + 1}{W^N} - \frac{j\Sigma(W^{2N} - 1)}{W^N} \\ &\quad + \sum_{i=1}^{N-1}\sum_{h=i+1}^{N} \Gamma_i \Gamma_h \left[W^{2N} - W^{N-2(h-i)} - W^{2(h-i)} + 1\right] = 0 \end{aligned} \tag{5}$$

where $\Sigma = \sum_{i=1}^{N} \Gamma_i$.

Given the low-coupling assumption, use an appropriate asymptotic expansion of $W^\pm = e^{2j(r\pi+\epsilon^\pm)/N} \approx e^{2jr\pi}\left(1 + 2j\epsilon^\pm/N\right)$ and consider Taylor expansion of the terms $W(\epsilon^\pm)$ and $\Gamma_i(\epsilon^\pm)$ in the dispersion relation at the second-order gives

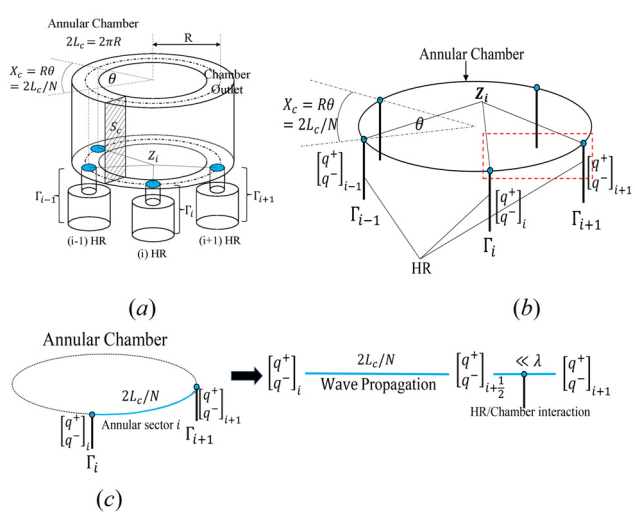


**Fig. 1 Schematic diagram of an HR-annular chamber system in (*a*) full 3D view, (*b*) a simplified one-dimensional network model, and (*c*) one module of the network model. Similar to diagrams from Refs. [21] and [22].**



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

$$A - 4B\epsilon^{\pm} + 4C\epsilon^{\pm 2} = 0 \tag{6}$$

where

$$\begin{aligned} A &= 4\sum_{i=1}^{N-1}\sum_{e=i+1}^{N}\Gamma_i^0\Gamma_e^0\left[\sin\left(\frac{2r\pi}{N}(e-i)\right)\right]^2 \\ B &= -\sum_{i=1}^{N-1}\Gamma_i^0 \\ C &= 1 \end{aligned} \tag{7}$$

thus $\epsilon^{\pm} = -(1/2)(\Sigma_0 \pm \varphi_0)$, where $\Sigma_0 = \sum_{i=1}^{N}\Gamma_i^0$ and $\varphi_0 = \sqrt{\Sigma_0^2 - A}$, and the 0 in $\Gamma_i^0$ represents the selected circumferential order designator. $\varphi_0$ is called splitting strength and can be expressed as

$$\begin{aligned} \varphi_0^2 &= \left[\sum_{i=1}^{N}\Gamma_i^0\cos(4r\pi i/N)\right]^2 + \left[\sum_{i=1}^{N}\Gamma_i^0\sin(4r\pi i/N)\right]^2 \\ &= \sum_{i=1}^{N}\Gamma_i^0 e^{j4r\pi i/N} \times \sum_{i=1}^{N}\Gamma_i^0 e^{-j4r\pi i/N} \end{aligned} \tag{8}$$

Then the eigenvalues of the system are (from the definition of $kL_c = r\pi + \epsilon^{\pm}$)

$$f^{\pm} = \frac{rc}{2L_c} + \frac{c}{2\pi L_c}\epsilon^{\pm} \tag{9}$$

The growth rate of the modes are

$$\mathrm{Im}(f^{\pm}) = -\frac{c}{4\pi L_c}\mathrm{Im}(\Sigma_0 \pm \varphi_0) \tag{10}$$

**2.2 The Effect of the Helmholtz Resonators' Distribution on Mode Splitting and Damping Properties.** Based on this theory, Yin and Yang [23] have studied the influence of the number of the same HRs which are (evenly distributed along the circumference) on the splitting of the first three circumferential modes. It was found that when the number of HRs is less than 7, the first three circumferential modes could be either degenerate or nondegenerate; when the number is greater than or equal to 7, the first three circumferential modes will all be degenerated. Following the same line, the annular chamber is uniformly divided into $N$ parts. However, we can either install or not install an HR for each part. For the position with HR placement, it is solved according to Eq. (2), and for the position without HR placement, it is considered to be the solid wall (i.e., $Z = \infty$, that is, $\Gamma = 0$). Therefore, the study will not be limited to the case of uniform HRs arrangement, and the previous study will be extended to the case of arbitrary HRs arrangement.

*2.2.1 The Splitting Strength With the Same Type of Helmholtz Resonators.* If only one type of HRs are considered ($\Gamma_i^0 = \Gamma^1$). Equation (8) can be written as

$$\begin{aligned} \varphi_0^2 &= \left[\sum_{i=1}^{N}\Gamma^1 e^{j4p\pi i/N}\right] \times \left[\sum_{i=1}^{N}\Gamma^1 e^{-j4r\pi i/N}\right] \\ &= \Gamma^1(a_1 + b_1 j) \times \Gamma^1(a_1 - b_1 j) \\ &= (\Gamma^1)^2(a_1^2 + b_1^2) \end{aligned} \tag{11}$$

We already know that the two solutions are

$$\begin{aligned} f^{+} &= \frac{rc}{2L_c} + \frac{c}{2\pi L_c}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right) \\ f^{-} &= \frac{rc}{2L_c} + \frac{c}{2\pi L_c}\left(-\frac{1}{2}\Sigma_0 - \frac{1}{2}\varphi_0\right) \end{aligned} \tag{12}$$

By combining Eqs. (2), (11), and (12), the follow relation can be obtained:

$$\begin{aligned} l_k &= \frac{\mathrm{Re}(f^{+}) - \mathrm{Re}(f^{-})}{\mathrm{Im}(f^{+}) - \mathrm{Im}(f^{-})} \\ &= \frac{\mathrm{Re}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right) - \mathrm{Re}\left(-\frac{1}{2}\Sigma_0 - \frac{1}{2}\varphi_0\right)}{\mathrm{Im}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right) - \mathrm{Im}\left(-\frac{1}{2}\Sigma_0 - \frac{1}{2}\varphi_0\right)} \\ &= \frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)} = \frac{\mathrm{Re}(\Gamma^1)}{\mathrm{Im}(\Gamma^1)} = \frac{\mathrm{Re}\left(-\frac{j}{2}\frac{S_i}{S_c Z}\right)}{\mathrm{Im}\left(-\frac{j}{2}\frac{S_i}{S_c Z}\right)} = \frac{\mathrm{Im}(Z)}{\mathrm{Re}(Z)} \end{aligned} \tag{13}$$

Equation (13) has an important indication: no matter what the number and the arrangement of the HRs are, the line linking the two split eigenvalues on the complex plane always have a fixed slope which is determined only by the impedance of the HR $Z$. Furthermore, the corresponding intercept $l_b = y - l_k x$, if we denote $y = \mathrm{Re}(f^{+})$ and $x = \mathrm{Im}(f^{+})$, and thus

$$\begin{aligned} l_b &= \mathrm{Re}\left(\frac{rc}{2L_c} + \frac{c}{2\pi L_c}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right)\right) \\ &\quad - \frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)} \times \mathrm{Im}\left(\frac{rc}{2L_c} + \frac{c}{2\pi L_c}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right)\right) \\ &= \mathrm{Re}\left(\frac{rc}{2L_c}\right) + \frac{c}{2\pi L_c} \times \mathrm{Re}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right) \\ &\quad - \frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)} \times \frac{c}{2\pi L_c} \times \mathrm{Im}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right) \end{aligned} \tag{14}$$

According to Eq. (11)

$$\begin{aligned} \varphi_0 &= \sqrt{\sum_{i=1}^{N}\Gamma^1 e^{j4r\pi i/N} \times \sum_{i=1}^{N}\Gamma^1 e^{-j4r\pi i/N}} \\ &= \Gamma^1 \times \sqrt{(a_1^2 + b_1^2)} \end{aligned} \tag{15}$$

Using this relationship and $\Sigma_0 = N \times \Gamma^1$, and Eq. (14) can be rewritten as

$$\begin{aligned} l_b &= \mathrm{Re}\left(\frac{rc}{2L_c}\right) + \frac{c}{2\pi L_c} \times \mathrm{Re}\left(-\frac{1}{2}N \times \Gamma^1 + \frac{1}{2}\sqrt{(a_1^2 + b_1^2)} \times \Gamma^1\right) \\ &\quad - \frac{\mathrm{Re}(N \times \Gamma^1)}{\mathrm{Im}(N \times \Gamma^1)} \times \frac{c}{2\pi L_c} \times \mathrm{Im}\left(-\frac{1}{2}N \times \Gamma^1 + \frac{1}{2}\sqrt{(a_1^2 + b_1^2)} \times \Gamma^1\right) \\ &= \mathrm{Re}\left(\frac{rc}{2L_c}\right) = f_0 \end{aligned} \tag{16}$$

We can now arrive at an important (and perhaps rather surprising) conclusion: when the same type of HRs are installed over the circumference, no matter how many HRs are used and no matter how they are distributed, the resulting eigenvalue solutions will always be on the same straight line

$$y = \frac{\mathrm{Im}(Z)}{\mathrm{Re}(Z)}x + f_0 \tag{17}$$

the slope is the division of the real part and the imaginary part of the impedance (acoustic impedance and acoustic resistance), and the intercept is the frequency of the acoustic mode without any resonator. $f_c = rc/2L_c + c/2\pi L_c(-(1/2)\Sigma_0)$ is the eigen value when the splitting strength is zero and we define it as the center point. Then the two solutions $f^{+}$ and $f^{-}$ when the splitting strength is not zero will appear in pairs along the line $y = (\mathrm{Im}(Z)/\mathrm{Re}(Z))x + f_0$ with $f_c$ as the center. A plot of this line and the corresponding eigenvalue solutions can be found in Sec. 3.1.3.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

*2.2.2 Considering Different Types of Helmholtz Resonators.* When more than one type of HRs are used, the result will change. We take the case with three types of HRs as an example to show how this could be analyzed. The three types of HRs are adjusted to have a resonant frequency close to the first-, second-, and third-order azimuthal modes, respectively. In this case, the splitting strength is

$$\begin{aligned}\varphi_0^2 &= \left(\sum_{q=1}^{N_1}\Gamma^1 e^{\frac{j4r\pi q_l}{N}} + \sum_{d=1}^{N_2}\Gamma^2 e^{\frac{j4r\pi d_l}{N}} + \sum_{t=1}^{N_3}\Gamma^3 e^{\frac{j4r\pi t_l}{N}}\right)\\ &\quad * \left(\sum_{i=1}^{N_1}\Gamma^1 e^{-\frac{j4r\pi q_l}{N}} + \sum_{m=1}^{N_2}\Gamma^2 e^{-\frac{j4r\pi d_l}{N}} + \sum_{n=1}^{N_3}\Gamma^3 e^{-\frac{j4r\pi t_l}{N}}\right)\\ &= \left(\Gamma^1(a_1+b_1 j) + \Gamma^2(a_2+b_2 j) + \Gamma^3(a_3+b_3 j)\right)\\ &\quad * \left(\Gamma^1(a_1-b_1 j) + \Gamma^2(a_2-b_2 j) + \Gamma^3(a_3-b_3 j)\right)\\ &= (\Gamma^1)^2(a_1^2+b_1^2) + (\Gamma^2)^2(a_2^2+b_2^2) + (\Gamma^3)^2(a_3^2+b_3^2)\\ &\quad + 2\Gamma^1\Gamma^2(a_1a_2+b_1b_2) + 2\Gamma^1\Gamma^3(a_1a_3+b_1b_3)\\ &\quad + 2\Gamma^2\Gamma^3(a_2a_3+b_2b_3)\end{aligned} \tag{18}$$

where $N = N_1 + N_2 + N_3$ and $q_l, d_l, t_l$ could be any integer between 1 and 24 ($q_l \neq d_l \neq t_l$).

The effect of only one type of HRs, $\Gamma^1$, cannot fully represent the overall effect. The other two types of HRs, $\Gamma^2$ and $\Gamma^3$, will inevitably affect the $\varphi_0$ value. However, if we're-interested in the influence of the HRs on the first azimuthal mode (our Taylor expansion is done around this mode), the impedance of the first type of HRs will be much smaller than those of the second and third types, resulting $|\Gamma^2| \ll |\Gamma^1|$ and $|\Gamma^3| \ll |\Gamma^1|$. Equation (18) can thus be approximately written as

$$\varphi_0^2 \approx (\Gamma^1)^2(a_1^2+b_1^2) \tag{19}$$

In this case, the influence of all the HRs is dominated by the arrangement of the first type of HRs, so the problem is the same as the case considering only one type of HRs $-f^+$ and $f^-$ are still distributed in pairs in straight lines.

If $|\Gamma^3| \ll |\Gamma^1|$, but $|\Gamma^2|$ is of the same order of magnitude as $|\Gamma^1|$—this happens when the resonant frequencies of the first and second types of HRs are not too far apart. Equation (18) can rewrited as

$$\begin{aligned}\varphi_0^2 &\approx (\Gamma^1)^2(a_1^2+b_1^2) + (\Gamma^2)^2(a_2^2+b_2^2)\\ &\quad + 2\Gamma^1\Gamma^2(a_1a_2+b_1b_2)\end{aligned} \tag{20}$$

At this point, it can be seen that the expression of the splitting strength is more complex and the distribution of the split modes on the complex plane is no longer along a straight line. For the first azimuthal mode ($r = 1$), if we assume that even though $\Gamma^2$ is not negligible, $|\Gamma^1|$ is significantly larger than $|\Gamma^2|$, so the term proportional to $(\Gamma^2)^2$ can be neglected. Using the same method as above, the slope $l_k$ is

$$\begin{aligned}l_k &= \frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)} = \frac{\mathrm{Re}\left((\Gamma^1)^2(a_1^2+b_1^2) + 2\Gamma^1\Gamma^2(a_1a_2+b_1b_2)\right)}{\mathrm{Im}\left((\Gamma^1)^2(a_1^2+b_1^2) + 2\Gamma^1\Gamma^2(a_1a_2+b_1b_2)\right)}\\ &= \frac{\mathrm{Re}\left(\left(\frac{1}{Z_l}\right)^2(a_1^2+b_1^2) + \frac{2}{Z_1Z_2}(a_1a_2+b_1b_2)\right)}{\mathrm{Im}\left(\left(\frac{1}{Z_l}\right)^2(a_1^2+b_1^2) + \frac{2}{Z_1Z_2}(a_1a_2+b_1b_2)\right)}\end{aligned} \tag{21}$$

the intercept $l_b$ is

$$\begin{aligned}l_b &= \mathrm{Re}\left(\frac{rc}{2L_c}\right) + \frac{c}{2\pi L_c}\times\mathrm{Re}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right)\\ &\quad - \frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)}\times\frac{c}{2\pi L_c}\times\mathrm{Im}\left(-\frac{1}{2}\Sigma_0 + \frac{1}{2}\varphi_0\right)\\ &= f_0 + \frac{c}{2\pi L_c}\times\mathrm{Re}\left(-\frac{1}{2}\Sigma_0\right)\\ &\quad - \frac{c}{2\pi L_c}\times\mathrm{Im}\left(-\frac{1}{2}\Sigma_0\right)\times\frac{\mathrm{Re}(\varphi_0)}{\mathrm{Im}(\varphi_0)}\end{aligned} \tag{22}$$

the center point $f_c$ is

$$f_c = \frac{rc}{2L_c} + \frac{c}{2\pi L_c}\left(-\frac{1}{2}\Sigma_0\right) \tag{23}$$

where $\Sigma_0 = N_1 \times \Gamma^1 + N_2 \times \Gamma^2$.

When the type and number of HRs are determined, $\Sigma_0$ and $Z_i$ are fixed. Therefore, the only factor that affects the splitting strength is the arrangements (In the above case, it is $a_1, b_1, a_2$, and $b_2$). $f^+$ and $f^-$ occur in pairs, and the values are distributed in two ranges. A plot of these eigenvalue solutions can be found in Sec. 3.1.4.

For an annular combustor with circumferential rotational symmetry and considering a linear flame transfer function, the circumferential thermoacoustic mode of the system is second-order degenerate in the absence of resonators. This means that the mode shape of the mode can involve any combinations of clockwise and counterclockwise pressure components, and is thus undetermined—can exhibit various mode properties, such as standing, spinning, or mixed modes. The introduction of resonators can break this rotational symmetry and the splitting strength is the key factor affecting the nature of the resultant modes. When the splitting strength is 0, the thermoacoustic mode maintains the second-order degenerate property, and when the splitting strength is nonzero, the mode shape is determined—these can be predicted by the present model.

*2.2.3 Analysis of the Best Damping Effect.* The effect of splitting strength on mode stability has been discussed in Ref. [21]. It was found that uniform distributions of the same type of burners could lead to mode splitting, and one of the two split modes become more stable but the other one is always more unstable. For the distribution of dampers, we can conclude that $f^\pm = f_c$ at $\varphi_0 = 0$; when $\varphi_0 \neq 0$: $f^\pm$ are two solutions distributed on both sides of the $f_c$. This is going to be discussed in Sec. 3.1.3. As the angle of inclination of the solution distribution line will not be equal to 90 deg ($\mathrm{Re}(Z) \neq 0$), when $\varphi_0 \neq 0$, there must have two solutions $f^\pm$ distributed along the distribution line on both sides of $f_c$. This means that there must be one solution that is more stable and the other more unstable. As the stability of the whole system is limited by the least stable mode, the less stable mode after considering a nonzero splitting strength is more likely to enter the unstable state compared to the case when the same $f_c$ but a lower (or zero for the optimum damping) splitting strength is achieved, as shown in Fig. 2. Therefore, the safest way is to directly find the solution with the minimum $\varphi_0$ so that the mode does not split or has minimum splitting, which can better ensure the suppression of oscillations. For the arrangement of a single type of resonator, the design scheme of $\varphi_0 = 0$ is the easiest and most effective. When there are multiple types of resonators, the distribution of mode solutions (three types are given in this paper) is going to be discussed in Sec. 3.1.5. However, satisfying the $\varphi_0 = 0$ requires more demanding conditions, such as the expression of the splitting intensity $\varphi_0$ derived from Eq. (18), and the effects of $a_1, a_2, a_3, b_1, b_2, b_3$ all need to be considered.

It should be clarified that zero splitting does not guarantee stability—it's determined by the acoustic energy balance between the source from the flame and the overall damping in the system.



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

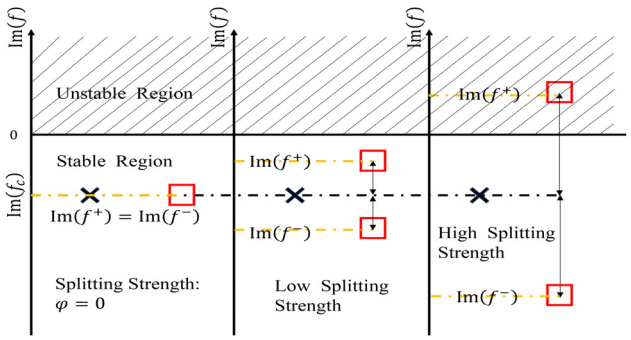


**Fig. 2 Stability of split modes under different splitting strengths**

Furthermore, under real geometry and operating conditions, there are many uncertainties that may need to be taken into account: nonlinear effect, noise from turbulent combustion, machining accuracy, etc. These effects could make it difficult to achieve the ideal state of $\varphi_0 = 0$. The present results could provide a theoretical guidance for further studies on these aspects.

## 3 Case Study and Discussions

We now consider an annular combustion chamber which is modeled as a one-dimensional annular duct. Its first three circumferential modes are calculated as 229 Hz, 458 Hz, and 687 Hz. According to the theoretical model, the term that causes mode splitting is $\varphi_0$, it is mainly related to the coupling parameter $\Gamma$. In the annular-combustor gas turbine of a certain type from Shanghai Electric, there are a total of 24 burners, and beside each burner one HR could be installed. In this real industrial gas turbine, three types of HRs have been installed, each tuned to one of the first three azimuthal modes. The number of each type of HRs being used is 7. In this paper, both seven and eight HRs for each type will be studied in order to find optimum HR distribution patterns.

**3.1 Considering 24 Helmholtz Resonators Categorized Into Three Groups.** We first consider the case with 24 HRs uniformly distributed over the circumference. These HRs consist of three types, each with eight HRs tuned to one of the first three azimuthal modes of the annular combustion chamber. When these HRs are installed, there are in total approximately $1.5 \times 10^9$ different distribution patterns. Using numerical simulations to obtain solutions of the targeted nonlinear eigenvalue system across all cases at this number is very heavy although not impossible, especially when the split modes are very close to each other so are very difficult to be accurately differentiated numerically. It should be noted that because the HRs are arranged in the circumferential direction, there must be two arrangements that conform to mirror symmetry or rotation axial symmetry, and the effects of these two arrangements are the same. That is to say: if the axis line is made at the position of 1–13, as shown in Fig. 3. Figures 3(*a*) and 3(*b*) are a mirror-symmetrical pair. On the other hand, due to the rotational symmetry of the system, when the system is rotated either clockwisely or anticlockwisely for an integer times of $2\pi/24$, the resultant distribution is the same as the original one, as shown in Figs. 3(*a*) and 3(*c*).

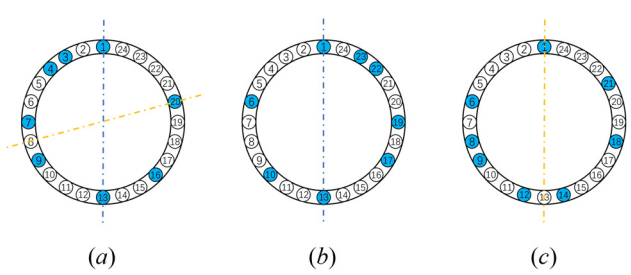


**Fig. 3 Schematic diagram of mirror symmetry and rotational symmetry. The highlighted blue circles denote the position of the resonators, and the other numbers correspond to hard walls ($Z = \infty$). (Color version online.)**

In the following part of this subsection, we will first study the case with eight HRs of only one type considering all their possible distribution patterns over the 24 positions. The corresponding mode splitting and damping properties will be discussed in detail. This will then be extended to consider all three types of HRs simultaneously. Based on the idea of the theoretical model, for the position where the HR is located, it is assumed that it has a corresponding impedance $Z_i$ and $\Gamma_i$ according to Eq. (2). Then the splitting strength $\varphi_0$ could be written as

$$\varphi_0^2 = \left(\sum_{q=1}^{N_1} \Gamma_i^0 e^{j4r\pi q_l/N}\right) \times \left(\sum_{q=1}^{N_1} \Gamma_i^0 e^{-j4r\pi q_l/N}\right) \tag{24}$$

The value of $N_1$ is 8, and $\Gamma_i^0$ is the value calculated based on the impedance at the targeted azimuthal mode frequency (because the HRs are all of the same type, it can be regarded as $\Gamma$). Among them, $q_l$ represents any eight of the 24 positions. For the analysis of this equation, we can start with discussing about the two extreme cases: the zero splitting case and the maximum splitting case.

*3.1.1 Eight Helmholtz Resonators of the Same Type and $\varphi_0 = 0$.* We now consider the first three azimuthal modes ($r = 1$, 2, 3) as an example to show how the problem could be studied. The values of $e^{j4\pi q_l/24}$ for $q_l = 1, 2, \ldots 24$ are shown on a complex plane in Fig. 4 when $q_l$ takes different values between 1 and 24, each $e^{j2r\pi q_l/24}$ gives a corresponding complex value (there will be a total of 24 complex values, but it is likely that some of them will be the same). We plot the corresponding complex values on the complex plane to get Fig. 4. And 12, 6, and 4 different positions (or values) over this unit circle would be obtained when $r$ is taken 1, 2, and 3, respectively. Note that $e^{-j4\pi q_l/24}$ is just the conjugate, so is not shown. The arrangements to achieve $\varphi_0 = 0$ can be considered from two perspectives.

The first way is to think from the perspective of the resultant force. The impact of each HR could be seen as a unit force directed from the origin point to the point of the HR on the unit circle. When the sum of

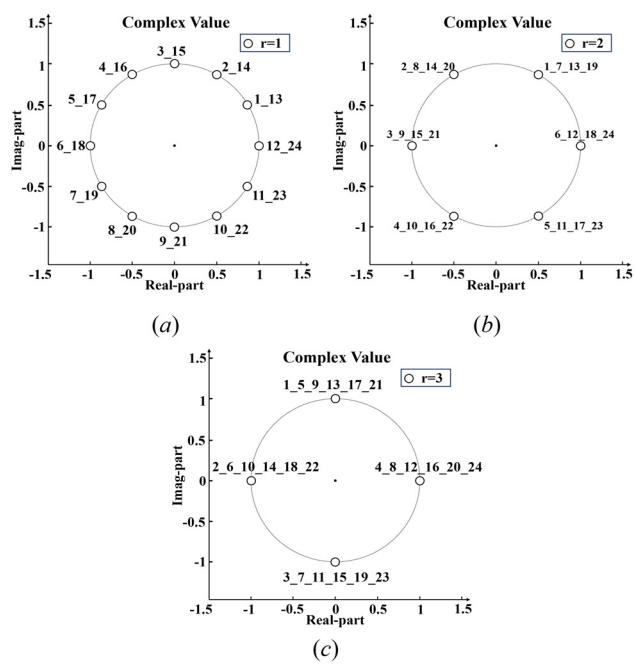


**Fig. 4 Complex values of $e^{\frac{j4\pi q_l}{24}}$ with $q_l = 1, 2, 3, \ldots, 24$. Different numbers represent complex values in different locations, the locations of the duplicates are separated by an underline.**



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

forces from all HRs is 0, the splitting strength $\varphi_0 = 0$. This can be proved by using Eq. (24). There are a number of options that can satisfy the above criteria.

For the $r = 1$ mode (Fig. 4(*a*)), this includes cases where forces are in all eight different directions (see Fig. 5(*b*)), six different directions (see Fig. 5(*c*)), and only four different directions (see Fig. 5(*a*)); for the $r = 2$ mode (as shown in Fig. 4(*b*)), this includes cases where the forces are in two directions (see Fig. 5(*d*)), four directions (see Figs. 5(*e*) and 5(*f*)); for the $r = 3$ mode (as shown in Fig. 4(*c*)), this includes cases where the forces are in four directions (see Figs. 5(*g*) and 5(*i*)) and two directions (see Fig. 5(*h*)).

In Fig. 5, the highlighted number denotes the position of the resonators, and the other numbers correspond to hard walls ($Z = \infty$). For example, Fig. 5(*a*) shows that we have placed resonators at eight positions with the highlighted numbers 1-4-7-10-13-16-19-22, and the arrow represents the force in the direction from the center of the circle to the position of each highlighted number, and the resultant overall force of all arrows is 0. Figure 5(*a*) corresponds to Fig. 6(*a*), which represents the placement of the resonator in the highlighted circles 1-4-7-10-13-16-19-22 and the rest of the hard walls ($Z = \infty$). Based on this, each of Figs. 5(*a*)–5(*i*) corresponds, respectively, to Figs. 6(*a*)–6(*i*) and they are all arrangements with a splitting strength of 0.

The second way is to think from a numerical perspective, $\sum_{q=1}^{N_1} \Gamma e^{\frac{j4r\pi q_l}{N}}$ and $\sum_{q=1}^{N_1} \Gamma e^{-\frac{j4r\pi q_l}{N}}$. The result is a complex number. If $\sum_{q=1}^{N_1} \Gamma e^{\frac{j4r\pi q_l}{N}} = G + Hj$, $\sum_{q=1}^{N_1} \Gamma e^{-\frac{j4r\pi q_l}{N}} = G - Hj$. The only way to make $\varphi_0 = 0$ is $(G + Hj) * (G - Hj) = G^2 + H^2 = 0$, that is, $G = H = 0$. Therefore all arrangements found in the complex domain satisfying this condition correspond to the cases in which the modes do not split.

The eight-HRs arrangements corresponding to cases in Fig. 5 are given, respectively, in Fig. 6. It is worth noting that these are all typical types of arrangements giving zero resultant force and thus zero splitting strength. These types of arrangements are important as it will be seen later that they provide the optimal damping performance among all possible arrangements of this eight-HRs system.

*3.1.2 Eight Helmholtz Resonators of the Same Type and $\varphi_0 = max$.* The arrangement with the largest splitting strength essentially corresponds to the situation where $\varphi_0 = \max$. Based on the idea that the resultant force can still be used, the resultant force of the eight selected HR positions in this case should be the one with the largest resultant force among all arrangements. Therefore, it is easy to think that the selected positions should avoid force cancelation between each other (even if there is, it should be as small as possible). Figure 7 shows the maximum resultant force for the first three modes. For example, Fig. 7(*a*) represents that we have placed the resonators in eight positions of the highlighted numbers 1-2-3-4-13-14-15-16, and the purple arrow represents the force in the direction from the center of the circle to the position of each resonator, and the overall resultant force is indicated by a red arrow (this is the maximum resultant force). The corresponding distribution patterns of the HRs are shown in Fig. 8. For the $r = 1$ mode, as two HRs at the opposite positions (with a $\pi$ angle in between) gives the same force, the opposite arrangement gives maximum overall force, so the optimum splitting was obtained with the distribution shown in Figs. 7(*a*) and 8(*a*). For the $r = 2$ and $r = 3$ modes, the rotation period for HRs to give the same force is reduced to $\pi/2$ and $\pi/3$, respectively. Therefore, it can be seen that the maximum splitting appears at the distribution patterns with HRs 1/4-circle and 1/6-circle apart, respectively, for $r = 2$ and $r = 3$. It should be noted that for the $r = 3$ case, only six HRs could be arranged 1/6-circle apart; the remaining two can only be placed at positions giving different forces compared to the six determined positions, but their resultant force should be as aligned as possible to that of the six given HRs, so two locations just beside the HRs already arranged are chosen.

*3.1.3 Eight Helmholtz Resonators of the Same Type for Arbitrary Arrangements.* Except for the two extreme cases discussed above, all other possible distribution patterns are now

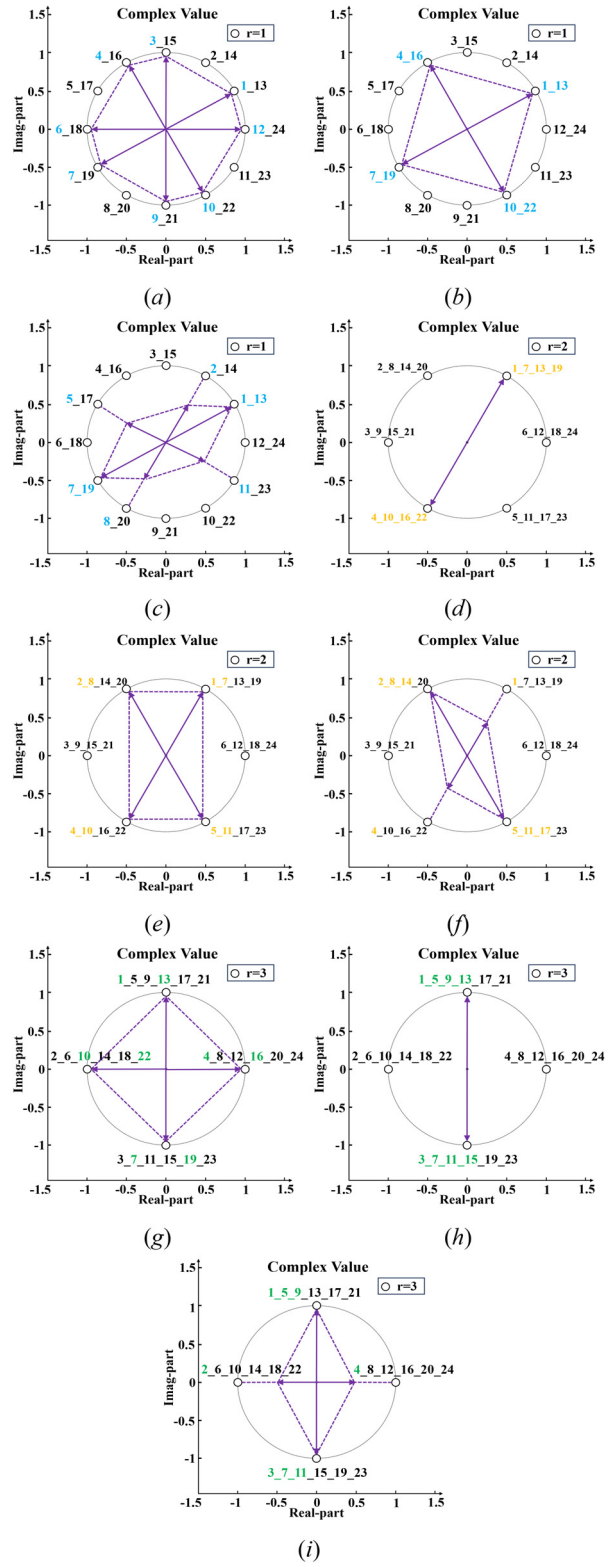


**Fig. 5 Schematic diagram of cases with a zero resultant overall force. The highlighted numbers represent the selected HR position, and the arrows represent the overall force in each direction.**

studied, and the results are shown in Fig. 9. It can be seen from the results that the two split solutions appear in pairs, and all solutions form a straight line with a specific slope and intercept. The function of this straight line is $y = (\mathrm{Im}(Z)/\mathrm{Re}(Z))x + f_0$, being consistent with our theoretical derivation in Sec. 2.2.1. It is interesting to see



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

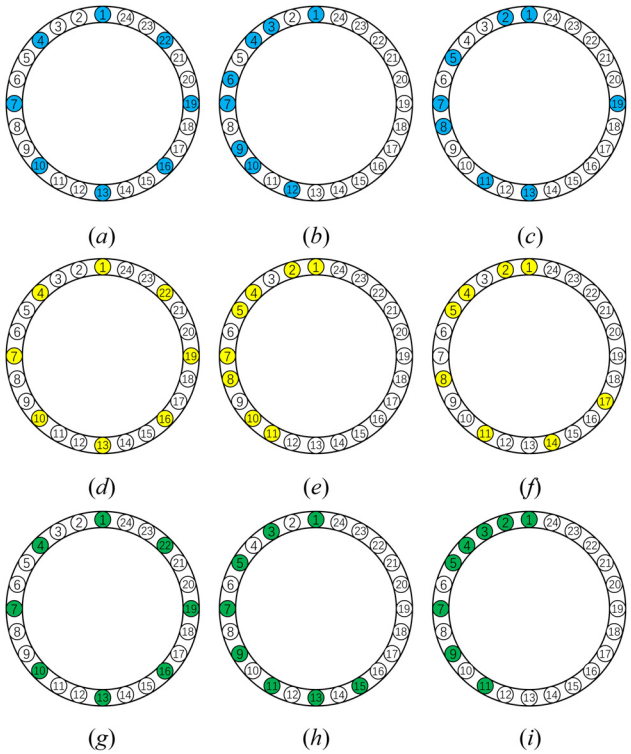


**Fig. 6 HR arrangement with a resultant force of 0. (*a*), (*b*), and (*c*) $r = 1$ mode; (*d*), (*e*), and (*f*) $r = 2$ mode; and (*g*), (*h*), and (*i*) $r = 3$ mode.**

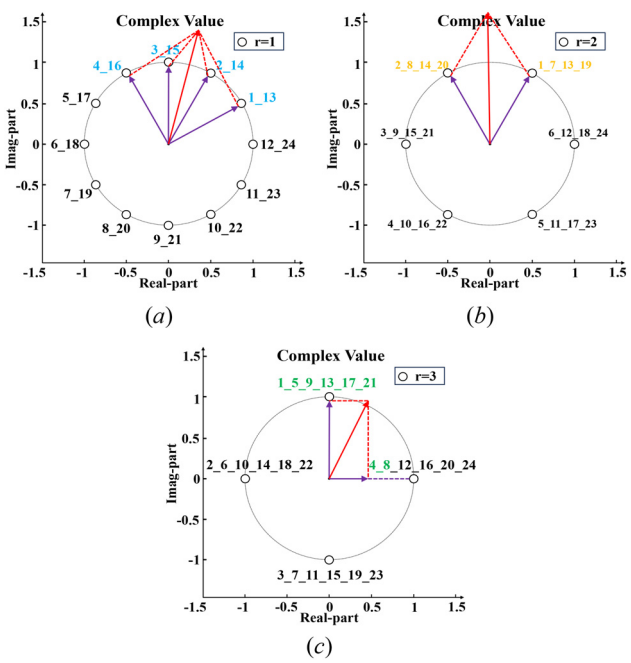


**Fig. 7 Schematic diagram of the maximum resultant force**

that although the number of the HRs' distribution patterns are the same for the $r = 1$, 2, and 3 cases, the number of different mode splitting solutions are much fewer for larger $rs$. This is because, with larger $rs$, more different HRs' distribution patterns could give the same splitting strength and thus the same mode solutions.

*3.1.4 The Effect of 24 Helmholtz Resonators With Three Different Types on a Particular Azimuthal Mode.* We now consider all three types of HRs, with the number of each type being 8. We first

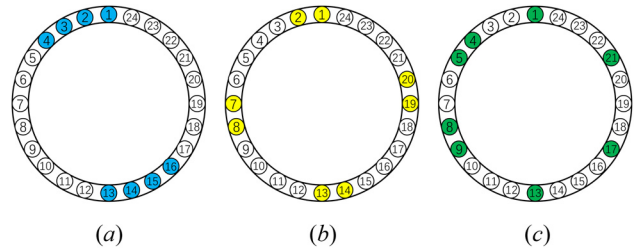


**Fig. 8 HR arrangement of the maximum resultant force (*a*) $r = 1$ mode; (*b*) $r = 2$ mode; and (*c*) $r = 3$ mode**

focus on only the effect on the first azimuthal mode. To this end, the expression of the splitting strength has been given in Eq. (18). The resonant frequency of the first type HR is tuned to match the first-order azimuthal mode of the annular combustion chamber, $f_1 = c/2/\pi\sqrt{S_1/l_1/V_1} = 286\,\text{Hz}$. For the second and third types of HRs, we first assume that their resonant frequencies are much higher than that of the first-order azimuthal mode, i.e., $f_2 = 2436\,\text{Hz}$, $f_3 = 2566\,\text{Hz}$. In this case, $|\Gamma^2| \ll |\Gamma^1|$ and $|\Gamma^3| \ll |\Gamma^1|$, so the first type HRs contribute dominantly to the influence on the first order azimuthal mode, and the results of the eigenvalue solution considering all HRs approximately overlap with the straight line predicted by Eq. (17), as shown in Fig. 10(*a*). Among all possible distribution patterns, Figs. 10(*b*) and 10(*c*) show one pattern corresponding to the maximum and minimum splitting strengths, respectively. It can be clearly seen that the maximum splitting corresponds to the case where the first eight HRs are split into two groups $\pi$-angle apart with each group having four neighboring HRs, being consistent with the maximum splitting pattern for the $r = 1$ mode shown in Fig. 8(*a*). The minimum splitting case is achieved with the eight first type HRs being uniformly distributed, corresponding to the first of the three cases shown in Fig. 8 where zero splitting has been achieved for the $r = 1$ mode. It is worth noting that, as $|\Gamma^2| \ll |\Gamma^1|$ and $|\Gamma^3| \ll |\Gamma^1|$, the detailed distributions of the second and third types of HRs changes very little of the results; they can be arbitrarily distributed on the rest positions.

If we change the resonant frequency of the second type HR to $f_2 = 398\,\text{Hz}$ and keep $f_3 = 2566\,\text{Hz}$, $|\Gamma^3| \ll |\Gamma^1|$ and $|\Gamma^3| \ll |\Gamma^2|$. Even though $|\Gamma^2|$ is still smaller than $|\Gamma^1|$, but they are now of the same order of magnitude. The eigenvalue solutions are shown in Fig. 11. It can be seen that the eigenvalue solutions are now scattered near the straight line of the case when only type one HR dominates, agreeing with our prediction in Sec. 2.2.2. It should be noted that for cases where none of these three types dominates, such as when $f_1$, $f_2$, and $f_3$ are close and the $r = 2$ mode is considered, a comprehensive consideration of all possible arrangements is needed.

*3.1.5 The Effect of 24 Helmholtz Resonators With Three Different Types on All of the First Three Azimuthal Modes.* Although the above case considers all the three types of HRs, it focuses on only one particular azimuthal mode. In a real gas turbine combustor, several azimuthal modes, such as the first three modes with $r = 1, 2, 3$, could all become unstable, so the design of multiple HRs may need to consider the optimum damping to all of these modes. Moreover, the frequencies of these modes may be not far away, so when each type of HRs are tuned to one azimuthal modes and the acoustic absorption bandwidth of the HRs are not too narrow, $\Gamma^1$, $\Gamma^2$, and $\Gamma^3$ could be of the same order of magnitude when targeting any one particular mode. Therefore, the key point to find optimum distributions of multiple HRs is to make sure that all targeted modes see as small as possible (ideally zero) splitting strength.

For the optimum damping of 24 HRs with three different types on all of the first three azimuthal modes, based on the analysis in Sec. 3.1.1 (particularly Figs. 6(*a*), 6(*d*), and 8(*g*)), one optimum design for achieving this is the uniform distribution of all the three types of HRs, as shown in Fig. 12. Although it might be a bit surprising to find

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

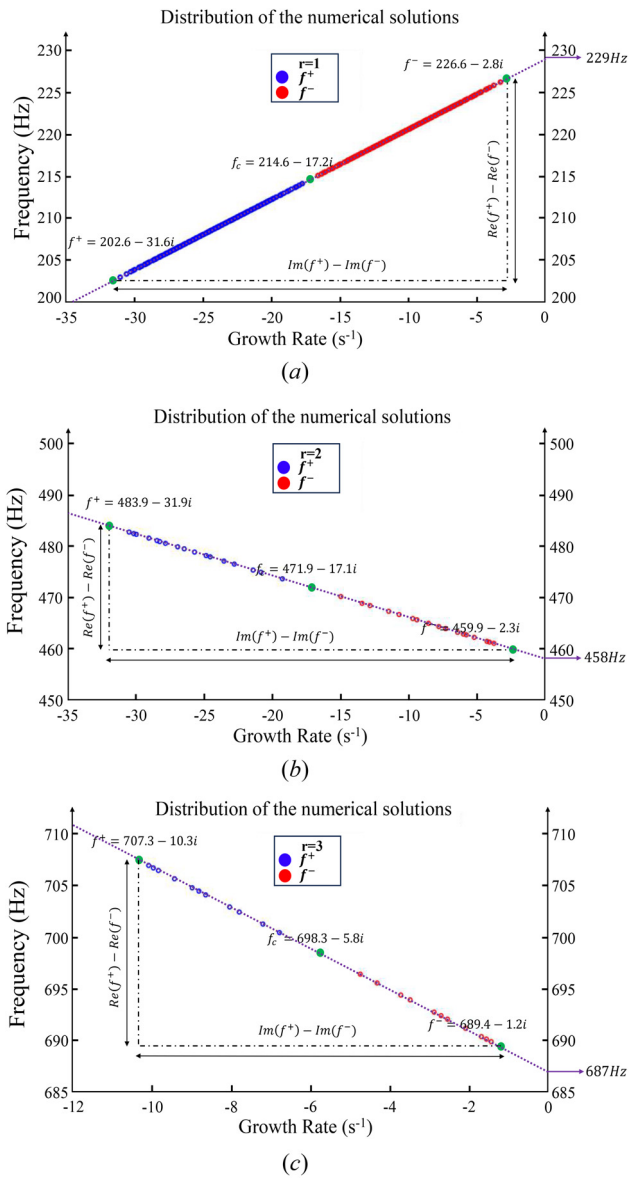


**Fig. 9 Frequency values of all arrangements, the $f^+$ and $f^-$ solutions correspond one-to-one. (Color version online.)**

that the optimum distribution is such a simple uniform (or rotationally symmetric) pattern, given the conclusion that for a given number of HRs, the optimum distribution is those giving zero splitting strength, it's very straightforward to understand this—for each type of resonator, this distribution pattern guarantees that the resultant force is 0 for any of the first three azimuthal modes.

**3.2 Considering 21 Helmholtz Resonators Categorized Into Three Groups.** The $24 = 8 + 8 + 8$ HRs case is special since all three types of HRs could be evenly distributed, and all of the first three azimuthal modes have zero splitting strength and thus optimum damping. In the real gas turbine from Shanghai Electric, 21 HRs with seven in each of the three types have been used. The number of burners and thus possible installation positions are still 24, so there will be three empty positions (without HR, so the impedance could be seen as infinitely large).

As shown in Figs. 13(*a*)–13(*c*), the distribution patterns to obtain the maximum splitting strength are similar to those for the case of eight HRs—the idea of maximizing the resultant force can still be used and the distribution patterns giving the maximum splitting are similar to those shown in Fig. 8. As for the minimum splitting, however, the results are quite different. The first difference is that

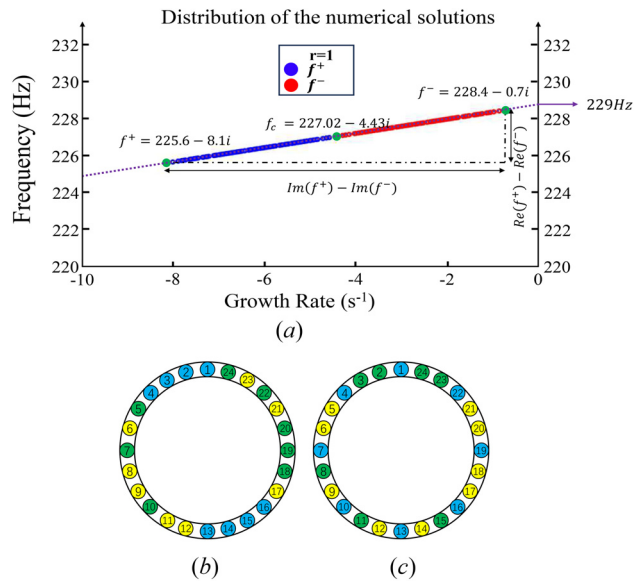


**Fig. 10 (*a*) Eigenvalue solutions of all possible HR distribution patterns, (*b*) the arrangement of maximum splitting, and (*c*) the arrangement of minimum splitting**

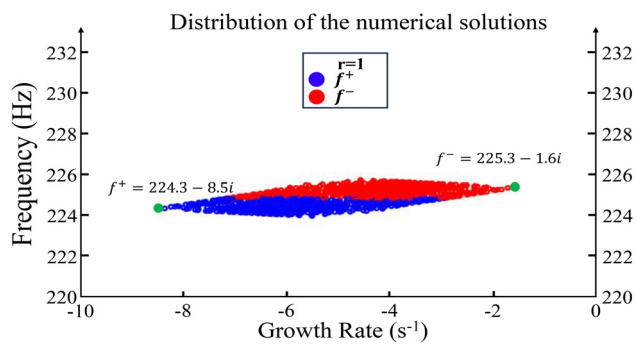


**Fig. 11 Schematic diagram of the distribution of solutions**

seven HRs cannot be evenly distributed now as seven is not a factor of 24. In addition, for the $r = 1$ and $r = 2$ cases, based on the rule of zero resultant force, it is still possible to find specific distribution patterns (such as those shown in Figs. 13(*d*) and 13(*e*), respectively) which give zero splitting. However, this is not possible anymore for the $r = 3$ case, so a pattern which gives the minimum (but not zero) splitting is given in Fig. 13(*f*).

When all three types of HRs are used and the damping on all three azimuthal modes are considered, two situations need to be considered. The first is the case when only one or two of the azimuthal modes require to be damped, i.e., the $r = 1$ and $r = 2$ modes. The first two types of HRs are tuned to resonant frequencies which are equal to these two modes, respectively. It is possible to

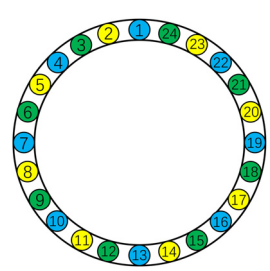

**Fig. 12 The arrangement of zero splitting strength $\varphi_0$=0 considering 24 HRs with three types**



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

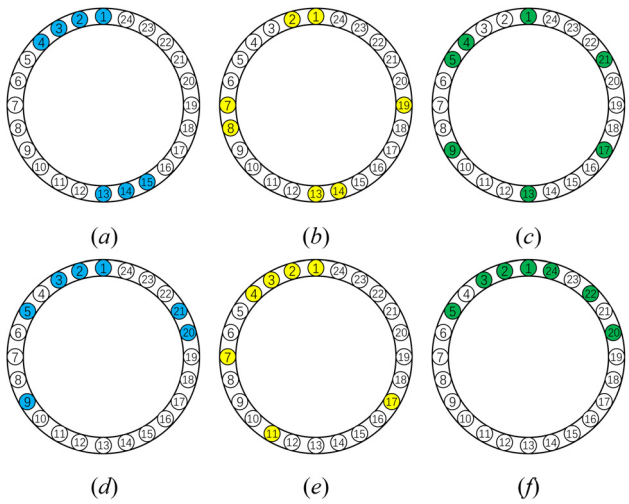


**Fig. 13 Arrangements of maximum splitting strength for $r = 1, 2, 3$ are shown in (*a*)–(*c*), respectively, and arrangements of minimum splitting strength for $r = 1, 2, 3$ are shown in (*d*)–(*f*), respectively**

prioritize the placement of these two types of HRs so they are both arranged in their optimum patterns. The third type HRs then have only ten positions left to choose—it is normally not possible to arrange this type in its optimum pattern (as that shown in Fig. 13(*f*)) so the best possible arrangement from these ten positions are used, as shown in Fig. 14(*a*). The second situation is that all three azimuthal modes could be unstable so all of them need to be considered in the optimum design. In this case, in order to find the optimum distribution pattern, an objective function is needed. This objective function could be the sum of the three splitting strengths, the growth rate of the least stable mode corresponding to all of the three azimuthal modes, or it could be an weighted sum of the growth rates of all modes, such as those done in Ref. [15]. When the objective is to minimize all three splitting strengths, the final optimum distribution pattern is usually the result of a compromise in the arrangement of the three types of HRs—it will satisfy the small overall splitting strength of the three modes, but it may not achieve the minimum splitting strength of any single mode. An optimum result of this case is shown in Fig. 14(*b*). Summarizing the results of this part, it can be found that to achieve an arrangement scheme with the best damping effect for multiple circumferential modes at the same time, it is necessary to have the minimum splitting strength (that is, the best damping effect) for each mode. When the HRs can be arranged evenly, this is generally a more direct solution; but once it is difficult to arrange evenly, not only a systematic calculation of Eq. (18) considering all possible arrangements is usually required, but also one or more carefully selected objective functions may need to be considered.

## 4 Application in a Real Industrial Gas Turbine

In the annular combustion chamber of a gas turbine from Shanghai Electric, thermoacoustic oscillations of the first three

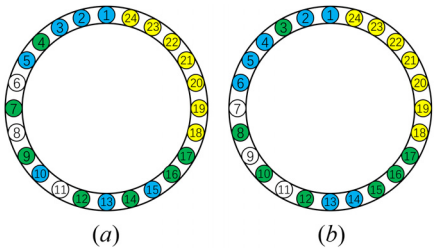


**Fig. 14 Arrangement of two different schemes (*a*) $r = 1$ and $r = 2$ modes are considered first, and the $r = 3$ mode is considered last. (*b*) Consider all three modalities and use the sum of the three splitting strengths as the objective function.**

azimuthal modes have been observed to occur and will affect the operation of the unit. Therefore, the circumferential HR arrangements studied in this paper are now used in a full 3D Helmholtz solver for the thermoacoustic instabilities of this combustor to assess their ability in damping thermoacoustic instabilities and provide suggestions on optimum distribution of these HRs.

**4.1 The Thermoacoustic Modes.** In this real annular combustor, thermoacoustic oscillations are more likely to occur in the circumferential mode at low frequencies [24]. Therefore, the circumferential modes without an HR is calculated (circumferential first-, second-, and third-order modes). The mean temperature field of the combustion system is obtained from a RANS simulation based on the $k - \epsilon$ turbulence and premixed combustion model. The mean temperature in both the plenum and the combustor match well with the measurements, guaranteeing the satisfactorily accurate estimation of the frequencies of the thermoacoustic modes. The flame transfer function is unknown, so a typical $n - \tau$ model with a two-dimensional scan of $n$ over [0, 1] and $\tau$ over [1 ms, 10 ms] has been performed, providing an approximate estimation (by comparing with onsite measurements of the dynamic pressure) of the unstable azimuthal modes with $n = 1$ and $\tau = 8, 8.2, 8.4, 8.6$ ms. As will be seen later, we consider four slightly different $\tau$s to illustrate the mode splitting property. The mode shapes of the first three unstable modes with $\tau = 8$ ms are shown in Figs. 15(*a*)–15(*c*). These three modes are the thermoacoustic modes that need to be suppressed, and thus the target of the design of the corresponding HRs.

**4.2 Damper Designs.** When the frequencies of the targeted modes are determined, we use the following empirical expression for resonant frequency to choose geometrical parameters for the HR: $f = c/2/\pi\sqrt{S/V/l}$, where $c$ is the speed of sound, $l$ and $S$ are the length and cross-sectional area of the HR's neck, respectively, and $V$ is the volume of the cavity. It should be noted that $V$ is limited by the space that can be used and $l$ limited by the thickness of the wall of the burner. The temperature and velocity of the mean flow passing through the HR neck are determined by those of the flow coming from the last stage of the compressor. After the geometrical and mean flow properties are determined, the impedance of the HR can be determined by using the same model as in Refs. [15] and [25]. In COMSOL, the HR is simplified to the equivalent impedance $Z$ and placed at the inlet of the combustion chamber, as shown in Fig. 15(*d*). Three different HR arrangements are considered: Arrangement one: the uniform distribution of $24 = 8 + 8 + 8$ HRs as shown in Fig. 12; Arrangement two: with the same $24 = 8 + 8 + 8$ HRs, but divide the circumference into three sectors, each with a central angle $2\pi/3$ and contains the same type of HR; Arrangement

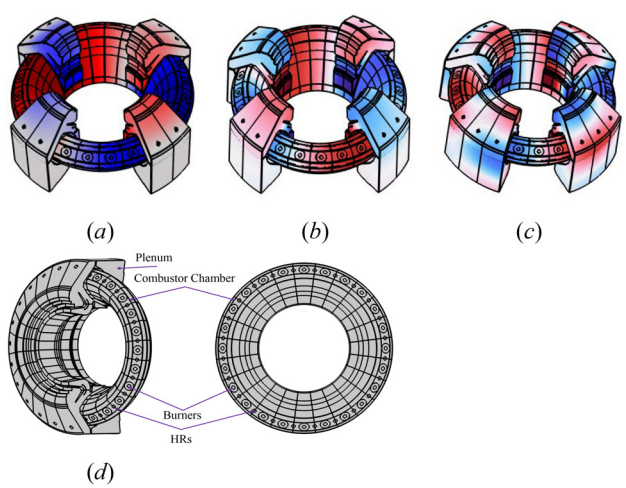


**Fig. 15 Circumferential first-, second-, and third-order acoustic modes of the annular combustion chamber**



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

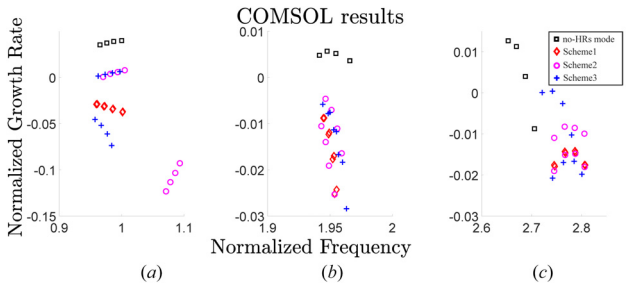


**Fig. 16 Splitting of the first three circumferential modes under different schemes. In the flame transfer function, $n = 1$ and $\tau = 8.0, 8.2, 8.4, 8.6$ms has been used to calculate each mode so four solutions were obtained for each mode [22].**

three: considering the $24 = 7 + 7 + 7 + 3$ arrangement as shown in Fig. 14(*b*)—with 21 HRs and three empty positions and optimizing the first three azimuthal modes simultaneously. It is expected that under the first arrangement scheme, the circumferential mode will not split, the second arrangement would be the arrangement with the largest mode splitting, and the third arrangement will give a smaller (but not zero) splitting compared to the second arrangement.

**4.3 Results and Discussions.** As shown in Fig. 16, without including any HR, the first three unstable modes are located at (1, 0.04), (1.966, 0.0036), and (2.706, −0.0087) ($\tau = 8$ ms has been considered and all results are normalized to the first mode). They correspond to the first three azimuthal modes of the combustion chamber, with the mode shapes shown in Figs. 15(*a*)–15(*c*). The black box represents the circumferential modes with no resonator added (the first-, second-, and third-order circumferential modes). After adding the resonator, the diamond mark represents the mode trajectory of scheme 1, the circle mark represents the mode trajectory of scheme 2, and the cross mark represents the mode trajectory of scheme 3. The reason that the frequencies of the third mode is lower than 3 is because the frequency is normalized by the frequency of the first mode, and the mode involve not just circumferential but also longitudinal wave propagations, and the effect of plenum-combustor coupling is not negligible.

With the HRs, no matter which arrangement is considered, all modes become more stable, showing the effective damping performance of the HRs. Considering the mode splitting effect, if we now use $\mathrm{abs}(|f^+ - f^-|)$ to quantify this, it can be clearly seen that the first arrangement scheme gives much smaller (nearly zero) splitting compared to that from schemes 2 and 3. Both schemes 2 and 3 give significant splitting results. Even though scheme 3 is supposed to be an optimal arrangement, the resulted mode splitting is not always negligible, especially for the first mode. This is because that an optimum design based on an objective function considering all modes does not guarantee that each mode will be well damped. It can be seen that the first scheme stabilized all of the three originally unstable azimuthal modes, while the second and third schemes both see two split modes, of which one mode could be unstable for the first and third azimuthal modes. This proves that the first scheme is a better design than the other two for the current case. The first distribution scheme results in no mode splitting while the other two both results in splitting and one mode less stable than the other, being consistent with our theoretical derivations in Sec. 2.

## 5 Conclusions

In this paper, we use a theoretical model and a 3D Helmholtz solver to study the effect of the distribution patterns of multiple HRs on the azimuthal thermoacoustic modes. Given the positions over the circumference on which the HRs can be installed, all possible circumferential arrangements of a single type of HRs, as well as two and three types of HRs each tuned to one of the first three azimuthal modes, are considered. It is shown that in realistic applications, such typical cases could involve a total number of possible arrangement patterns as large as $\mathcal{O}(10^9)$. Based on the analysis of the splitting strength and the optimal damping effect, the optimal arrangement schemes targeting the first-order, second-order, and third-order circumferential modes are obtained. The arrangement schemes are applied to the annular combustion chamber of a gas turbine from Shanghai Electric, and the 3D FEM Helmholtz solver from COMSOL is used to calculate the effect of installing the designed arrangement patterns of HRs and compare with the theory. The main findings are listed below.

(1) When the same type of HRs are considered, and if we use $f^{\pm}$ to denote the two modes after splitting, it was found that, no matter how many HRs are used and no matter how they are arranged over the circumference when their number is fixed, the two split modes will always distribute along a given straight line with $f_c$ as the center. $f_c = rc/2L_c + c/2\pi L_c(-(1/2)\Sigma_0)$ depends on the number and type of HRs, and $f^+$ and $f^-$ are related to the splitting strength $\varphi_0$. The slope of the straight line is determined by the resonator impedance $Z$ as $\mathrm{Im}(Z)/\mathrm{Re}(Z)$, and the intercept is the frequency $f_0$ of the circumferential mode without HR. $\Sigma_0 = \sum_{i=1}^{N}\Gamma_i$ depends on the type and number of HRs, and determines the optimum damping.

(2) The splitting strength $\varphi_0$ can be understood by using the concept of the overall resultant force: the splitting strength is the largest when the resultant force is maximum (That is $|f^+ - f^-| = \max$). When the resultant force is 0, the splitting strength is 0. The optimal damping effect is obtained when the splitting strength is 0 ($f^+ = f^- = f_c$).

(3) When there are multiple types of HRs, each tuned to target a certain mode, such as the first three azimuthal modes, the relation of these modes need to be considered. When a particular target mode is far away from the other modes, only the HRs that are tuned to this particular target mode needs to be considered when calculating the effect on the target mode. In this case, $f^{\pm}$ are still distributed along a given straight line no matter how the HRs are arranged. If the target modes are close to each other, all types of HR need to be considered, and the calculation of all arrangements is necessary.

(4) The idea of overall resultant force can still be used to obtain the optimal distribution patterns of multitype HRs: the net force of each type of HR for each target mode is 0. When the resultant force cannot be exactly 0 or multiple targeted modes need to be considered simultaneously, it is usually necessary to select an appropriate objective function, i.e., prioritizing one or two modes, or optimizing the overall damping to all modes.

(5) Even in the 3D simulation of a realistic complex combustor, the first scheme given in Fig. 16 was found being able to keep the modes not split (or with a very small splitting) thus provides an optimum damping. A random arrangement would usually split the modes, leading to one mode less damped than that of the optimum case.

## Acknowledgment

The authors would like to thank the National Natural Science Foundation of China (Project No. 52106159) and Science Center for Gas Turbine Project (P2022-B-II-013-001) for supporting this research.

## Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

## Nomenclature

### Latin Characters

$c$ = speed of sound (m s$^{-1}$)
$f$ = frequency (Hz)



Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026

$k$ = wave number
$L$ = circumference (m)
$N$ = total number of HRs
$p$ = pressure (Pa)
$r$ = circumferential mode order
$S$ = cross-sectional area ($m^2$)
$V$ = HR cavity volume ($m^2$)
$Z$ = acoustic impedance ($Pa \cdot s\ m^{-3}$)

### Greek Symbols

$\Gamma$ = coupling parameter
$\epsilon$ = wavenumber perturbation
$\rho$ = density ($kg\ m^{-3}$)
$\Sigma$ = coupling strength
$\varphi$ = splitting strength

### Subscripts

$c$ = combustor section
$i$ = Helmholtz resonator section

### Superscript

$'$ = fluctuating quantity

## References


[1] Dowling, A. P., and Stow, S. R., 2003, "Acoustic Analysis of Gas Turbine Combustors," J. Propul. Power, **19**(5), pp. 751–764.
[2] O'Connor, J., Acharya, V., and Lieuwen, T., 2015, "Transverse Combustion Instabilities: Acoustic, Fluid Mechanic, and Flame Processes," Prog. Energy Combust. Sci., **49**, pp. 1–39.
[3] Poinsot, T., 2017, "Prediction and Control of Combustion Instabilities in Real Engines," Proc. Combust. Inst., **36**(1), pp. 1–28.
[4] Prieur, K., Durox, D., Schuller, T., and Candel, S., 2017, "A Hysteresis Phenomenon Leading to Spinning or Standing Azimuthal Instabilities in an Annular Combustor," Combust. Flame, **175**, pp. 283–291.
[5] Zhao, D., and Li, X., 2015, "A Review of Acoustic Dampers Applied to Combustion Chambers in Aerospace Industry," Prog. Aerosp. Sci., **74**, pp. 114–130.
[6] Gysling, D. L., Copeland, G. S., McCormick, D. C., and Proscia, W. M., 2000, "Combustion System Damping Augmentation With Helmholtz Resonators," ASME J. Eng. Gas Turbines Power, **122**(2), pp. 269–274.
[7] Dupère, I. D. J., and Dowling, A. P., 2005, "The Use of Helmholtz Resonators in a Practical Combustor," ASME J. Eng. Gas Turbines Power, **127**(2), pp. 268–275.
[8] Schulze, M., Kathan, R., and Sattelmayer, T., 2015, "Impact of Absorber Ring Position and Cavity Length on Acoustic Damping," J. Spacecr. Rockets, **52**(3), pp. 917–927.
[9] Bothien, M., Noiray, N., and Schuermans, B., 2014, "A Novel Damping Device for Broadband Attenuation of Low-Frequency Combustion Pulsations in Gas Turbines," ASME J. Eng. Gas Turbines Power, **136**(4), p. 041504.
[10] Yang, D., and Morgans, A. S., 2017, "Acoustic Models for Cooled Helmholtz Resonators," AIAA J., **55**(9), pp. 3120–3127.
[11] Betz, M., Zahn, M., Hirsch, C., and Sattelmayer, T., 2019, "Impact of Damper Placement on the Stability Margin of an Annular Combustor Test Rig," ASME Paper No. GT2019-90238.
[12] Lepers, J., Krebs, W., Prade, B., Flohr, P., Pollarolo, G., and Ferrante, A., 2005, "Investigation of Thermoacoustic Stability Limits of an Annular Gas Turbine Combustor Test-Rig With and Without Helmholtz-Resonators," ASME Paper No. GT2005-68246.
[13] Zahn, M., Betz, M., Schulze, M., Hirsch, C., and Sattelmayer, T., 2017, "Predicting the Influence of Damping Devices on the Stability Margin of an Annular Combustor," ASME Paper No. GT2017-64238.
[14] Mensah, G. A., and Moeck, J. P., 2017, "Acoustic Damper Placement and Tuning for Annular Combustors: An Adjoint-Based Optimization Study," ASME J. Eng. Gas Turbines Power, **139**(6), p. 061501.
[15] Yang, D., Sogaro, F. M., Morgans, A. S., and Schmid, P. J., 2019, "Optimising the Acoustic Damping of Multiple Helmholtz Resonators Attached to a Thin Annular Duct," J. Sound Vib., **444**, pp. 69–84.
[16] Stow, S. R., and Dowling, A. P., 2003, "Modelling of Circumferential Modal Coupling Due to Helmholtz Resonators," ASME Paper No. GT2003-38168.
[17] Noiray, N., Bothien, M., and Schuermans, B., 2011, "Investigation of Azimuthal Staging Concepts in Annular Gas Turbines," Combust. Theory Modell., **15**(5), pp. 585–606.
[18] Noiray, N., and Schuermans, B., 2013, "On the Dynamic Nature of Azimuthal Thermoacoustic Modes in Annular Gas Turbine Combustion Chambers," Proc. R. Soc. A: Math., Phys. Eng. Sci., **469**(2151), p. 20120535.
[19] Mazur, M., Nygård, H. T., Dawson, J., and Worth, N., 2018, "Experimental Study of Damper Position on Instabilities in an Annular Combustor," ASME Paper No. GT2018-75070.
[20] Humbert, S. C., Moeck, J. P., Paschereit, C. O., and Orchini, A., 2023, "Symmetry-Breaking in Thermoacoustics: Theory and Experimental Validation on an Annular System With Electroacoustic Feedback," J. Sound Vib., **548**, p. 117376.
[21] Bauerheim, M., Salas, P., Nicoud, F., and Poinsot, T., 2014, "Symmetry Breaking of Azimuthal Thermo-Acoustic Modes in Annular Cavities: A Theoretical Study," J. Fluid Mech., **760**, pp. 431–465.
[22] Li, C., Yang, D., Li, S., and Zhu, M., 2019, "An Analytical Study of the Effect of Flame Response to Simultaneous Axial and Transverse Perturbations on Azimuthal Thermoacoustic Modes in Annular Combustors," Proc. Combust. Inst., **37**(4), pp. 5279–5287.
[23] Yin, L., and Yang, D., 2023, "Modal Coupling and Mode Degeneracy in Annular Combustor With Multiple Helmholtz Resonators: A Theoretical Study," Symposium on Thermoacoustics in Combustion: Industry Meets Academia (SoTiC 2023), Zurich, Switzerland, Sept. 11–14, p. 72.
[24] Vignat, G., Durox, D., Prieur, K., and Candel, S., 2019, "An Experimental Study Into the Effect of Injector Pressure Loss on Self-Sustained Combustion Instabilities in a Swirled Spray Burner," Proc. Combust. Inst., **37**(4), pp. 5205–5213.
[25] Howe, M. S., and Lighthill, M. J., 1979, "On the Theory of Unsteady High Reynolds Number Flow Through a Circular Aperture," Proc. R. Soc. London, Ser. A, **366**(1725), pp. 205–223.


Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/147/2/021004/7383084/gtp_147_02_021004.pdf by guest on 01 January 2026